

PROGRAM 12:-

AIM

To implement the Tic-Tac-Toe game using Python and allow two players to play the game by placing X and O alternatively.

ALGORITHM

1. Start the program.
2. Create a 3×3 board with empty spaces.
3. Assign X to Player 1 and O to Player 2.
4. Display the current board.
5. Ask the current player to select a position.
6. Place the player's symbol in the selected position.
7. Check whether the player has three symbols in a row, column, or diagonal.
8. If a player wins, display the winner.
9. If all positions are filled without a winner, declare a draw.
10. Repeat until the game ends.
11. Stop the program.

PROGRAM

```
board = [' ' for _ in range(9)]

def display():
    print(board[0], '|', board[1], '|', board[2])
    print('--+---+--')
    print(board[3], '|', board[4], '|', board[5])
    print('--+---+--')
    print(board[6], '|', board[7], '|', board[8])

def check_win(player):
    wins = [(0,1,2),(3,4,5),(6,7,8),(0,3,6),(1,4,7),(2,5,8),(0,4,8),(2,4,6)]
    return any(board[a] == board[b] == board[c] == player for a,b,c in wins)

player = 'X'

for turn in range(9):
```

```
display()

pos = int(input("Enter position (1-9): ")) - 1

if board[pos] == ' ':
    board[pos] = player
    if check_win(player):
        display()
        print("Player", player, "wins!")
        break
    player = 'O' if player == 'X' else 'X'
else:
    print("Position already occupied!")
else:
    display()
    print("Game Draw!")
```

OUTPUT

```

  |   |
--+---+--
  |   |
--+---+--
  |   |
Enter position (1-9): 5
  |   |
--+---+--
  | x |
--+---+--
  |   |
Enter position (1-9): 1
O |   |
--+---+--
  | x |
--+---+--
  |   |
Enter position (1-9): 9
O |   |
--+---+--
  | x |
--+---+--
  |   | x
Enter position (1-9): 7
O |   |
--+---+--
  | x |
--+---+--
O |   | x
Enter position (1-9): 3
O |   | x
--+---+--
  | x |
--+---+--
O |   | x
Enter position (1-9): 4
O |   | x
--+---+--
O | x |
--+---+--
O |   | x
Player O wins!

```

RESULT

The Tic-Tac-Toe game was successfully implemented, and the winner was determined based on the positions selected by the players.